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3

BASIC KINETIC THEORY

3.1 The velocity distribution functions

A gas flow would be completely described, in a classical sense, by listings of the position, velocity, and internal state of every molecule at a particular instant. The number of molecules in a real gas is so large that such a description is unthinkable, and we must resort to a statistical description in terms of probability distributions. A number of distinct velocity distribution functions are employed in kinetic theory and possible confusion may be avoided by a general review of the relationships between them.

We will commence by defining the *single particle distribution function in velocity space*. Consider a sample of gas that is homogeneous in physical space and contains N identical molecules. A typical molecule has a velocity $\boldsymbol{c}$ with components u, v, and w in the direction of the Cartesian axes x, y, and z. Just as x, y, and z define a space called physical space, u, v, and w define *velocity space*, as shown in Fig. 3.1. Each molecule can be represented in this space by the point defined by its velocity vector. The velocity distribution function $f(\boldsymbol{c})$ is then defined by

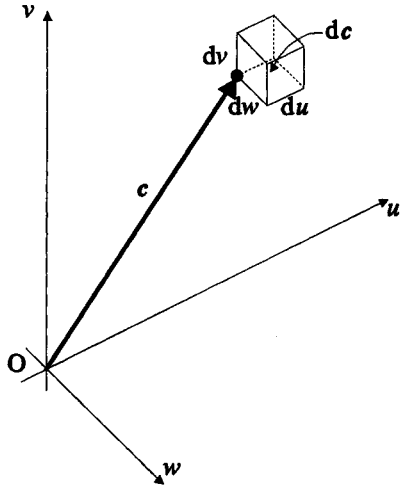


Fig. 3.1 Typical molecule and element in velocity space.

$$dN = Nf(\boldsymbol{c})\, du\, dv\, dw, \tag{3.1}$$

where dN is the number of molecules in the sample with velocity components u to $u+du$, v to $v+dv$, and w to $w+dw$. The product $du\, dv\, dw$ may be identified as a volume element in velocity space and is denoted by $d\boldsymbol{c}$. An alternative form of eqn (3.1) is, therefore,

$$dN = Nf(\boldsymbol{c})\, d\boldsymbol{c}, \tag{3.1a}$$

and this need not be restricted to Cartesian coordinates. The functional statement is usually omitted so that $f(\boldsymbol{c})$ becomes simply f. Also, since both dN and N refer to the molecules in the same volume of physical space, the number density may be used in place of the number. Therefore, the fraction of molecules within the velocity space element $d\boldsymbol{c}$ is

$$dn/n = f\, d\boldsymbol{c}. \tag{3.1b}$$

Since every molecule is represented by a point in velocity space,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f\, du\, dv\, dw = \int_{-\infty}^{\infty} f\, d\boldsymbol{c} = N/N = 1. \tag{3.2}$$

This distribution function is therefore a normalized function such that its integration over all velocity space yields unity. Note that f can never be negative and must either have finite bounds in velocity space or tend to zero as c tends to infinity.

The macroscopic quantities were defined in Chapter 1 in terms of averages over the molecular velocities. These averages may be established as instantaneous, time, or ensemble averages over the molecules in an element of physical space. These molecules may be regarded as constituting a homogeneous gas sample and the single particle distribution function in velocity space provides an appropriate description. In order to relate the macroscopic properties to this distribution function, we must determine the relationship between the function and the average value of any molecular quantity Q. This quantity is either a constant or a function of the molecular velocity. The mean value principle gives

$$\bar{Q} = \frac{1}{N}\int_N Q\, dN,$$

and substituting for dN from eqn (3.1a),

$$\bar{Q} = \frac{1}{N}\int_{-\infty}^{\infty} Q f(\boldsymbol{c}) N\, d\boldsymbol{c}.$$

Omitting the functional statements, we have the general result

$$\bar{Q} = \int_{-\infty}^{\infty} Q f\, d\boldsymbol{c}. \tag{3.3}$$

This process is often referred to as establishing a *moment* of the distribution function and the macroscopic properties are referred to as moments of the distribution function. For example, the stream velocity that was defined by eqn (1.21) as $\bar{\boldsymbol{c}}$ may be written

$$\bar{\boldsymbol{c}} = \int_{-\infty}^{\infty} \boldsymbol{c} f \, \mathrm{d}\boldsymbol{c}. \tag{3.4}$$

The macroscopic flow properties are generally functions of position and time, and it is sometimes desirable to express the explicit dependence of the distribution function on the position vector in velocity space $\boldsymbol{r}$, and the time t. Just as $\mathrm{d}\boldsymbol{c}$ has been used to denote a volume element in velocity space, a volume element in physical space may be denoted by $\mathrm{d}\boldsymbol{r}$. The product $\mathrm{d}\boldsymbol{c}\mathrm{d}\boldsymbol{r}$ then denotes a volume element in *phase space*, which is the multi-dimensional space formed by the combination of physical space and velocity space. The *single particle distribution function in phase space* $\mathscr{F}(\boldsymbol{c},\boldsymbol{r},t)$ is defined by

$$\mathrm{d}N = \mathscr{F}(\boldsymbol{c},\boldsymbol{r},t)\, \mathrm{d}\boldsymbol{c}\, \mathrm{d}\boldsymbol{r}, \tag{3.5}$$

with $\mathrm{d}N$ now representing the number of molecules in the phase space element $\mathrm{d}\boldsymbol{c}\mathrm{d}\boldsymbol{r}$. In Cartesian coordinates, $\mathrm{d}\boldsymbol{c}\mathrm{d}\boldsymbol{r}$ becomes $\mathrm{d}u\mathrm{d}v\mathrm{d}w\mathrm{d}x\mathrm{d}y\mathrm{d}z$ and $\mathrm{d}N$ is the number of molecules with velocity components ranging from u to $u+\mathrm{d}u$, v to $v+\mathrm{d}v$, and w to $w+\mathrm{d}w$ and spatial coordinates ranging from x to $x+\mathrm{d}x$, y to $y+\mathrm{d}y$, and z to $z+\mathrm{d}z$. Note that $\mathscr{F}$ defines the number rather than the fraction of molecules in the phase space element. It has not been normalized and an integration over the whole phase space yields the total number of molecules in the system N, rather than unity.

If the distribution function in velocity space $f(\boldsymbol{c})$ is applied to the physical space element $\mathrm{d}\boldsymbol{r}$, the number of molecules N in eqn (3.1) represents the total number of molecules in $\mathrm{d}\boldsymbol{r}$, while $\mathrm{d}N$ represents the number of molecules in the phase space element $\mathrm{d}\boldsymbol{c}\mathrm{d}\boldsymbol{r}$. We can then write

$$\mathrm{d}N = N f(\boldsymbol{c})\, \mathrm{d}\boldsymbol{c} = \mathscr{F}(\boldsymbol{c},\boldsymbol{r},t)\, \mathrm{d}\boldsymbol{c}\, \mathrm{d}\boldsymbol{r}$$

and, since the number density in the phase space element is $N/\mathrm{d}\boldsymbol{r}$,

$$n f(\boldsymbol{c}) = \mathscr{F}(\boldsymbol{c},\boldsymbol{r},t).$$

Therefore, when $f(\boldsymbol{c})$ is used in a context in which it also depends on $\boldsymbol{r}$ and t, we have

$$nf \equiv \mathscr{F}. \tag{3.6}$$

We will take advantage of this identity to use f exclusively in the chapters that follow. Some authors (for example, Chapman and Cowling (1952) and Harris (1971)) have preferred $\mathscr{F}$, which they have denoted by f. Therefore, in common with a number of other authors (for example Kennard 1938 and Vincenti and Kruger 1965), our equations in nf will be equivalent to their equations in f.

The most specific distribution function is that for all N molecules in the system. At any instant, a complete system of monatomic molecules can be represented by a point in $6N$ dimensional phase space. If we consider a large number or ensemble of such systems, the probability of finding a system in the volume element $d\boldsymbol{c}_1 d\boldsymbol{c}_2 \ldots d\boldsymbol{c}_N d\boldsymbol{r}_1 d\boldsymbol{r}_2 \ldots d\boldsymbol{r}_N$ about the phase space point $\boldsymbol{c}_1, \boldsymbol{r}_1, \boldsymbol{c}_2, \boldsymbol{r}_2, \ldots, \boldsymbol{c}_N, \boldsymbol{r}_N$ is

$$F^{(N)}(\boldsymbol{c}_1, \boldsymbol{r}_1, \boldsymbol{c}_2, \boldsymbol{r}_2, \ldots, \boldsymbol{c}_N, \boldsymbol{r}_N, t) d\boldsymbol{c}_1 d\boldsymbol{c}_2 \ldots d\boldsymbol{c}_N \, d\boldsymbol{r}_1 d\boldsymbol{r}_2 \ldots d\boldsymbol{r}_N , \tag{3.7}$$

thus defining the N *particle distribution function* $F^{(N)}$. The subscript denotes the number of the molecule. A *reduced distribution function* $F^{(R)}$ for R of the N molecules is defined by

$$F^{(R)}(\boldsymbol{c}_1, \boldsymbol{r}_1, \boldsymbol{c}_2, \boldsymbol{r}_2, \ldots, \boldsymbol{c}_R, \boldsymbol{r}_R, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F^{(N)} d\boldsymbol{c}_{R+1} \ldots d\boldsymbol{c}_N d\boldsymbol{r}_{R+1} \ldots d\boldsymbol{r}_N . \tag{3.8}$$

In particular, the *single particle distribution function* $F^{(1)}(\boldsymbol{c}_1, \boldsymbol{r}_1, t)$ is obtained by setting R=1. The probability of finding molecule number 1 in the phase space element $d\boldsymbol{c}_1 d\boldsymbol{r}_1$ at time t is $F^{(1)}(\boldsymbol{c}_1, \boldsymbol{r}_1, t)$ irrespective of the positions of the other $N-1$ molecules. Since the molecules are indistinguishable, the number of molecules in the phase space element at time t is $NF^{(1)}$. We therefore have

$$NF^{(1)} \equiv \mathscr{F} \tag{3.9}$$

and $F^{(1)}$ can be regarded as the normalized version of $\mathscr{F}$.

The two particle distribution function $F^{(2)}(\boldsymbol{c}_1, \boldsymbol{r}_1, \boldsymbol{c}_2, \boldsymbol{r}_2, t)$ is of particular importance when considering binary collisions. The definition of a dilute gas requires that only a very small fraction of the space occupied by the gas actually contains a molecule. Therefore, in such a gas, it is generally assumed that the probability of finding a pair of molecules in a particular two particle configuration is simply the product of the probabilities of finding the individual molecules in the two corresponding one particle configurations. This requires

$$F^{(2)}(\boldsymbol{c}_1, \boldsymbol{r}_1, \boldsymbol{c}_2, \boldsymbol{r}_2, t) = F^{(1)}(\boldsymbol{c}_1, \boldsymbol{r}_1, t) F^{(1)}(\boldsymbol{c}_2, \boldsymbol{r}_2, t) \tag{3.10}$$

and expresses the principle of *molecular chaos*. While the higher-order distribution functions are required for the study of dense gases, the single particle distribution function provides an adequate description of dilute gases.

If the molecules are diatomic or polyatomic, the dimensions of phase space are increased by the number of internal degrees of freedom. Also, if the molecules are not spherically symmetric, their orientations must be specified. In general, the dimensions of phase space are equal to the least number of scalar variables that are required to specify the position, velocity, orientation, and internal state of a molecule. Separate distribution functions are required for each species of a gas mixture. It is hardly surprising that most presentations of kinetic theory deal almost exclusively with gases consisting of a single species of monatomic molecules.

3.2 The Boltzmann equation

We have seen that the velocity distribution functions provide a statistical description of a gas on the molecular level. The next step is to establish the relationships between the distribution functions and the variables on which they depend. Ideally, the resulting equations would allow analytical solutions of problems in molecular gas dynamics.

The basic statistical mechanics equation for a gas is the Liouville equation which expresses the conservation of the N particle distribution function in $6N$ dimensional phase space. This equation is not directly useful, since the description of a real gas flow in terms of $F^{(N)}$ is completely out of the question. However, just as a hierarchy of reduced distribution functions $F^{(R)}$ was defined by eqn (3.8), a hierarchy of equations called the BBGKY equations may be obtained through the repeated integration of the Liouville equation. The final equation in the hierarchy is for the single particle distribution function $F^{(1)}$ and is the only one to hold out some hope of solution for flows of engineering interest. This equation also involves the two particle distribution function $F^{(2)}$, but becomes a closed equation for $F^{(1)}$ when molecular chaos (eqn (3.10)) is assumed. Then, through eqn (3.9), this becomes an equation for the single particle distribution in phase space, and is equivalent to the equation that was originally formulated by Boltzmann (1872). The mathematical limits that define the validity of the Boltzmann equation are most precisely established through the derivation from the Liouville equation (see, for example: Grad (1958); Cercignani (1969); or Harris (1971)). On the other hand, the physical significance of each term in the equation is more readily appreciated if the single particle distribution function is used throughout a derivation from first principles. The latter procedure will be followed here and, for clarity and simplicity, the derivation will be restricted to a simple gas.

At a particular instant, the number of molecules in the phase space element $\mathrm{d}\boldsymbol{c}\,\mathrm{d}\boldsymbol{r}$ is given by eqn (3.5) as $\mathscr{F}\mathrm{d}\boldsymbol{c}\,\mathrm{d}\boldsymbol{r}$. The identity (3.6) enables this to be written $nf\mathrm{d}\boldsymbol{c}\,\mathrm{d}\boldsymbol{r}$. If the location and shape of the element does not vary with time, the rate of change of the number of molecules in the element is

$$\frac{\partial}{\partial t}(nf)\,\mathrm{d}\boldsymbol{c}\,\mathrm{d}\boldsymbol{r}. \tag{3.11}$$

The processes that contribute to the change in the number of molecules within $\mathrm{d}\boldsymbol{c}\,\mathrm{d}\boldsymbol{r}$ are illustrated in Fig. 3.2. They are:

(i) The convection of molecules across the face of $\mathrm{d}\boldsymbol{r}$ by the molecular velocity $\boldsymbol{c}$. The representation of the phase space element as separate volume elements in physical and velocity space emphasizes the fact that $\boldsymbol{c}$ and $\boldsymbol{r}$ are treated as independent variables. $\boldsymbol{c}$ is regarded as a constant within $\mathrm{d}\boldsymbol{r}$, and $\mathrm{d}\boldsymbol{c}$ is regarded as being located at the point defined by $\boldsymbol{r}$.

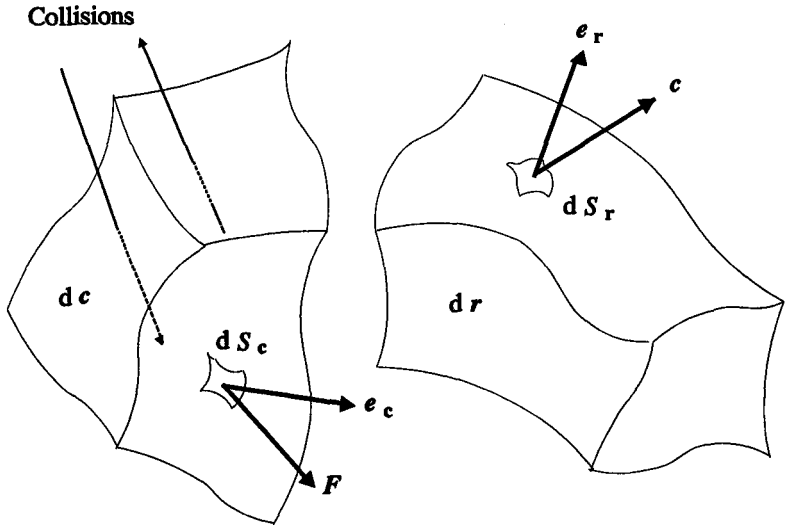


Fig. 3.2 Molecular flux to and from a phase space element $d\boldsymbol{c}d\boldsymbol{r}$.

(*ii*) The 'convection' of molecules across the surface of $d\boldsymbol{c}$ as a result of the external force per unit mass $\boldsymbol{F}$. The effect of the acceleration $\boldsymbol{F}$ on the molecules in $d\boldsymbol{c}$ is analogous to the effect of the velocity $\boldsymbol{c}$ on the molecules in $d\boldsymbol{r}$.

(*iii*) The scattering of molecules into and out of $d\boldsymbol{c}d\boldsymbol{r}$ as a result of intermolecular collisions. The gas is assumed to be dilute, as defined by eqn (1.9) and discussed in §1.3. One consequence of this assumption is that a collision may be assumed to be an instantaneous event at a fixed location in physical space. This means that a collision causes a molecule to jump from one point to another in velocity space, but it remains at the same point in physical space and time. Therefore, in Fig. 3.2, collisions are represented as affecting only the element $d\boldsymbol{c}$. A second major consequence of the dilute gas assumption is that all collisions may be assumed to be binary collisions.

First consider process (*i*) which is a conservative process across the surface $d\boldsymbol{r}$. The number of molecules in the phase space element is $nfd\boldsymbol{c}d\boldsymbol{r}$, so the number density of class $\boldsymbol{c}$ molecules within $d\boldsymbol{r}$ is $nfd\boldsymbol{c}$. Eqns (1.17) and (3.3) then enable the net inflow of molecules of this class across the surface of $d\boldsymbol{r}$ to be written as

$$-\int_{S_r} nf\boldsymbol{c}.\boldsymbol{e}_r \, dS_r \, d\boldsymbol{c}.$$

Here, S_r is the total area of the surface of $d\boldsymbol{r}$, dS_r is an element of this surface, and $\boldsymbol{e}_r$ is the unit normal vector of this element. Gauss' theorem enables the surface integral over S_r to be converted to a volume integral over $d\boldsymbol{r}$. The expression then becomes

$$-\int_{d\boldsymbol{r}} \nabla.(nf\boldsymbol{c})\, d(d\boldsymbol{r})\, d\boldsymbol{c}$$

or, since nf and $\boldsymbol{c}$ are constants within $d\boldsymbol{r}$,

$$-\nabla.(nf\boldsymbol{c})\, d\boldsymbol{r}\, d\boldsymbol{c}.$$

Also, since we are considering only molecules of class $\boldsymbol{c}$, the velocity $\boldsymbol{c}$ may be taken outside the divergence in physical space. Therefore, the inflow of molecules of class $\boldsymbol{c}$ across the surface of $d\boldsymbol{r}$ due to the velocity $\boldsymbol{c}$ is

$$-\boldsymbol{c}.\frac{\partial(nf)}{\partial \boldsymbol{r}}\, d\boldsymbol{c}\, d\boldsymbol{r}. \tag{3.12}$$

We may take advantage of the analogy between process (*ii*) in velocity space and process (*i*) in physical space to write the inflow of molecules across the surface of $d\boldsymbol{c}$, due to the external force per unit mass $\boldsymbol{F}$, as

$$-\boldsymbol{F}.\frac{\partial(nf)}{\partial \boldsymbol{c}}\, d\boldsymbol{c}\, d\boldsymbol{r}. \tag{3.13}$$

The total number of molecules scattered out of the element $d\boldsymbol{c}d\boldsymbol{r}$ as a result of collisions is readily obtained through an analysis similar to that which led to eqns (1.10) and (1.11) for the collision frequency and total number of collisions in the gas. However, in order to obtain a meaningful expression for the molecules scattered into the element, we must consider both the pre-collision and post-collision velocities of the molecules participating in the collisions. In particular, we are concerned with the collision of a molecule of class $\boldsymbol{c}$ with one of class $\boldsymbol{c}_1$ such that their post-collision velocities are $\boldsymbol{c}^*$ and $\boldsymbol{c}_1^*$, respectively. This is called a class $\boldsymbol{c}, \boldsymbol{c}_1 \rightarrow \boldsymbol{c}^*, \boldsymbol{c}_1^*$ collision, and we will now calculate the rate of scattering of molecules of class $\boldsymbol{c}$ out of $d\boldsymbol{c}d\boldsymbol{r}$ as a result of collisions of this class. A molecule of class $\boldsymbol{c}$ may be chosen as a test particle moving with speed c_r among stationary field molecules of class $\boldsymbol{c}_1$. The volume swept out in physical space by the cross-section for this class of collision is $c_r\sigma d\Omega$ and the number of class $\boldsymbol{c}_1$ molecules per unit volume in physical space is $nf_1 d\boldsymbol{c}_1$. The number of collisions of this class suffered by the test molecule per unit time is, therefore,

$$nf_1 c_r \sigma\, d\Omega\, d\boldsymbol{c}_1.$$

Since the number of class $\boldsymbol{c}$ molecules in the phase space element is $nfd\boldsymbol{c}d\boldsymbol{r}$, the number of class $\boldsymbol{c}, \boldsymbol{c}_1 \rightarrow \boldsymbol{c}^*, \boldsymbol{c}_1^*$ collisions per unit time in the element is

$$n^2 f f_1 c_r \sigma\, d\Omega\, d\boldsymbol{c}_1\, d\boldsymbol{c}\, d\boldsymbol{r}. \tag{3.14}$$

Just as f denotes the value of the velocity distribution function f at $\boldsymbol{c}$, f_1 denotes the value of f at $\boldsymbol{c}_1$. Similarly, f^* and f_1^* may be used to denote the values of f at $\boldsymbol{c}^*$ and $\boldsymbol{c}_1^*$, respectively. Note also that the expression of a binary collision probability in terms of the product of two single particle distribution functions has implicitly invoked the principle of molecular chaos.

The existence of inverse collisions (Fig. 2.2) means that an analysis, exactly similar to that leading to eqn (3.14) may be made for the collisions of class $\boldsymbol{c}, \boldsymbol{c}_1 \leftarrow \boldsymbol{c}^*, \boldsymbol{c}_1^*$ that scatter molecules into class $\boldsymbol{c}$. This yields

$$n^2 f^* f_1^* c_r^* (\sigma \, \mathrm{d}\Omega)^* \, \mathrm{d}\boldsymbol{c}_1^* \, \mathrm{d}\boldsymbol{c}^* \, \mathrm{d}\boldsymbol{r} \tag{3.15}$$

for the collision rate in the phase space element $\mathrm{d}\boldsymbol{c}^* \mathrm{d}\boldsymbol{r}$. Eqn (2.8) shows that c_r^* is equal to c_r, while the symmetry between the direct and inverse collisions is such that there is a unit Jacobian for the transformation between the pre-collision and post-collision values of the product of the differential cross-section and velocity space elements. That is,

$$|(\sigma \, \mathrm{d}\Omega) \, \mathrm{d}\boldsymbol{c}_1 \, \mathrm{d}\boldsymbol{c}| = |(\sigma \mathrm{d}\Omega)^* \, \mathrm{d}\boldsymbol{c}_1^* \, \mathrm{d}\boldsymbol{c}^*| \tag{3.16}$$

and eqn (3.15) may be written

$$n^2 f^* f_1^* c_r \sigma \, \mathrm{d}\Omega \, \mathrm{d}\boldsymbol{c}_1 \, \mathrm{d}\boldsymbol{c} \, \mathrm{d}\boldsymbol{r}. \tag{3.17}$$

The rate of increase of molecules of class $\boldsymbol{c}$ in the phase space element $\mathrm{d}\boldsymbol{c}\mathrm{d}\boldsymbol{r}$ as a result of the combined direct and inverse collisions of class $\boldsymbol{c}, \boldsymbol{c}_1 \leftrightarrow \boldsymbol{c}^*, \boldsymbol{c}_1^*$ is obtained by subtracting the loss rate (expression (3.14)) from the rate of gain (expression (3.17)). This gives

$$n^2 (f^* f_1^* - f f_1) c_r \sigma \, \mathrm{d}\Omega \, \mathrm{d}\boldsymbol{c}_1 \, \mathrm{d}\boldsymbol{c} \, \mathrm{d}\boldsymbol{r}. \tag{3.18}$$

The total rate of increase of molecules of class $\boldsymbol{c}$ in the element as a result of collisions is given by the integration of this expression over the complete cross-section for its collision with class $\boldsymbol{c}_1$ molecules, followed by the integration of the class $\boldsymbol{c}_1$ over all velocity space. The required expression for process (*iii*) is, therefore,

$$\int_{-\infty}^{\infty} \int_{0}^{4\pi} n^2 (f^* f_1^* - f f_1) c_r \sigma \, \mathrm{d}\Omega \, \mathrm{d}\boldsymbol{c}_1 \, \mathrm{d}\boldsymbol{c} \, \mathrm{d}\boldsymbol{r}. \tag{3.19}$$

The expression (3.11) for the total rate of increase of molecules of class $\boldsymbol{c}$ due to all three processes may be equated to the sum of expression (3.19) for process (*iii*) and expressions (3.12) and (3.13) for processes (*i*) and (*ii*), respectively. If the latter terms are transferred to the left-hand side and the complete equation is divided by $\mathrm{d}\boldsymbol{c}\mathrm{d}\boldsymbol{r}$, we have the *Boltzmann equation* for a simple dilute gas. This is

$$\frac{\partial}{\partial t}(nf) + \boldsymbol{c} \cdot \frac{\partial}{\partial \boldsymbol{r}}(nf) + \boldsymbol{F} \cdot \frac{\partial}{\partial \boldsymbol{c}}(nf) = \int_{-\infty}^{\infty} \int_{0}^{4\pi} n^2 (f^* f_1^* - f f_1) c_r \sigma \, \mathrm{d}\Omega \, \mathrm{d}\boldsymbol{c}_1. \tag{3.20}$$

In a gas mixture consisting of a total of s separate species, a separate distribution function must be defined for each species. The Boltzmann equation then becomes a set of s simultaneous equations and, as in §1.5, particular species may be represented by the subscripts p or q. The Boltzmann equation for species p of the mixture can therefore be written

$$\frac{\partial}{\partial t}(n_p f_p) + \boldsymbol{c}_p \cdot \frac{\partial}{\partial \boldsymbol{r}}(n_p f_p) + \boldsymbol{F} \cdot \frac{\partial}{\partial \boldsymbol{c}}(n_p f_p) = \sum_{q=1}^{s} \int_{-\infty}^{\infty} \int_{0}^{4\pi} n_p n_q (f_p^* f_{1q}^* - f_p f_{1q}) c_{rpq} \sigma_{pq} \, \mathrm{d}\Omega \, \mathrm{d}\boldsymbol{c}_{1q}. \tag{3.21}$$

It was noted in the preceding section that the presence of internal degrees of freedom requires the definition of extended distribution functions that allow for the additional dimensions of phase space. Also, the collision cross-sections of asymmetric molecules are a function of the molecular orientation and therefore change with time as the molecules rotate between collisions. Moreover inverse collisions do not exist for the classical models of polyatomic molecules. It is, however, possible to define cross-sections that are smoothed or averaged over the molecular rotations and vibrations. The Liouville theorem then leads to a unit Jacobean for the transformation corresponding to eqn (3.16), thus permitting a formulation similar to eqn (3.20). Chapman and Cowling (1970) have called the resulting equation the *generalized Boltzmann equation*.

The term on the right-hand side of the Boltzmann equation is called the *collision term*. Its integral form contrasts with the partial differential form of the terms expressing space and time dependence of nf, and is responsible for much of the mathematical difficulty associated with the Boltzmann equation. On the other hand, nf is the only dependent variable in the equation. This might be considered an advantage when comparing the Boltzmann equation with the Navier–Stokes equations of continuum gas dynamics, since these have the velocity components and, allowing for the equation of state, two of the thermodynamic properties as dependent variables. However, this advantage is far outweighed by the addition of the velocity space coordinates to the list of independent variables. A homogeneous gas problem becomes one-dimensional because the velocity distribution function is spherically symmetric in velocity space. This distribution becomes axially symmetric for a one-dimensional problem in physical space, so that a Boltzmann solution becomes a three-dimensional problem. For two- and three-dimensional flows there are no symmetries in velocity space and the problems become five- or six-dimensional. Time is an additional dimension if the problem is unsteady.

There is no possibility of obtaining analytical solutions of the Boltzmann equation for flows that involve complex geometries or large disturbances. Numerical solutions that require a computational mesh in phase space pose formidable problems because of the number of dimensions and the difficulties in setting the bounds of velocity space.

3.3 The moment and conservation equations

The quantity Q relates to a single molecule and is either a constant or a function of the molecular velocity. We have previously seen that the average value of this quantity may be obtained through the multiplication of the velocity distribution function by Q, followed by the integration of the product over all velocity space. These averages are referred to as moments of the distribution function. Similarly, a *moment of the Boltzmann equation* may be obtained by multiplying it by the quantity Q and then integrating the resulting equation over all velocity space. Since the moments of the distribution function include all the macroscopic properties of a monatomic gas, the moment equations may be expected to include the monatomic gas version of the conservation equations of continuum gas dynamics.

The multiplication of the Boltzmann equation (3.20) by Q yields

$$Q\frac{\partial}{\partial t}(nf)+Q\boldsymbol{c}.\frac{\partial}{\partial \boldsymbol{r}}(nf)+Q\boldsymbol{F}.\frac{\partial}{\partial \boldsymbol{c}}(nf) = Q\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2(f^*f_1^*-ff_1)c_r\sigma\,\mathrm{d}\Omega\,\mathrm{d}\boldsymbol{c}_1. \tag{3.22}$$

Both f and Q refer to molecules of class $\boldsymbol{c}$ and the moment equation is obtained by integrating over all classes of molecule. Since Q is either a constant or a function of $\boldsymbol{c}$ only, it may be taken within the derivative in the first term. The required integral of this term is, therefore,

$$\int_{-\infty}^{\infty}\frac{\partial}{\partial t}(nQf)\mathrm{d}\boldsymbol{c}$$

or, using eqn (3.3),

$$\frac{\partial}{\partial t}(n\bar{Q}). \tag{3.23}$$

Both $\boldsymbol{c}$ *and* Q may be taken inside the derivative in the second term of eqn (3.22), the integral becoming

$$\int_{-\infty}^{\infty}\nabla.(n\boldsymbol{c}Qf)\,\mathrm{d}\boldsymbol{c}.$$

or

$$\nabla.(n\overline{\boldsymbol{c}Q}). \tag{3.24}$$

The reason why $\boldsymbol{c}$ may be taken inside the derivative in eqn (3.22) is that the dependent variable in the Boltzmann equation is the value of the distribution function for this particular class of molecule. Therefore, while f is a function of $\boldsymbol{r}$ and t, $\boldsymbol{c}$ and Q may be regarded as being independent of $\boldsymbol{r}$ and t in eqn (3.22). However, the averages of $\boldsymbol{c}$ and Q are established

through the distribution function f and must be treated as functions of $\boldsymbol{r}$ and t. The integral in the third term in eqn (3.22) is

$$\int_{-\infty}^{\infty} \boldsymbol{Q}\boldsymbol{F} \cdot \frac{\partial}{\partial \boldsymbol{c}} (nf)\, \mathrm{d}\boldsymbol{c}\,,$$

which may be written

$$\int_{-\infty}^{\infty} \boldsymbol{F} \cdot \frac{\partial}{\partial \boldsymbol{c}} \; (n\boldsymbol{Q}f)\, \mathrm{d}\boldsymbol{c} - \int_{-\infty}^{\infty} \boldsymbol{F} \frac{\partial \boldsymbol{Q}}{\partial \boldsymbol{c}} nf\, \mathrm{d}\boldsymbol{c}.$$

It is assumed that $\boldsymbol{F}$ is independent of $\boldsymbol{c}$ and, since $f = 0$ or $f \to 0$ as $c \to \infty$, the first integral vanishes and the second becomes

$$-n\boldsymbol{F} \cdot \overline{\frac{\partial \boldsymbol{Q}}{\partial \boldsymbol{c}}}. \tag{3.25}$$

The integral of the term on the right-hand side of eqn (3.22) is called the *collision integral* and is denoted by $\Delta[Q]$, i.e.

$$\Delta\,[Q] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 Q(f^* f_1^* - f f_1) c_r \sigma\, \mathrm{d}\Omega\, \mathrm{d}\boldsymbol{c}_1\, \mathrm{d}\boldsymbol{c}\,. \tag{3.26}$$

Eqns (3.23) – (3.26) may now be brought together to write the moment equation for Q as

$$\frac{\partial}{\partial t}(n\bar{Q}) + \nabla \cdot (n\overline{\boldsymbol{c}Q}) - n\boldsymbol{F} \cdot \overline{\frac{\partial Q}{\partial \boldsymbol{c}}} = \Delta\,[Q]\,. \tag{3.27}$$

This equation may also be called the *transfer equation* or the *equation of change*.

Two symmetries are associated with the collision term and they lead to several alternative forms of $\Delta[Q]$. These clarify the physical meaning of this term and will be required for future applications. The first symmetry is between the collision partners and means that $\Delta[Q]$ is unchanged if $\boldsymbol{c}$ and $\boldsymbol{c}_1$ are interchanged and Q, which represents the value of Q at $\boldsymbol{c}$, is interchanged with the value Q_1 at $\boldsymbol{c}_1$. Similarly, $\boldsymbol{c}^*$ and Q^* may be interchanged with $\boldsymbol{c}_1^*$ and Q_1^*. The second symmetry is based in the existence of inverse collisions and is between the pre-collision and post-collision velocities. It enables Q and Q^* or Q_1 and Q_1^* to be interchanged, as long as $\boldsymbol{c}_1$, $\boldsymbol{c}$, f_1, and f are interchanged with $\boldsymbol{c}_1^*$, $\boldsymbol{c}^*$, f_1^*, and f^*. Advantage is taken of eqn (3.16) to avoid replacing $\mathrm{d}\boldsymbol{c}\,\mathrm{d}\boldsymbol{c}_1$ by $\mathrm{d}\boldsymbol{c}^*\,\mathrm{d}\boldsymbol{c}_1^*$. Application of the first symmetry to eqn (3.26) yields

$$\Delta\,[Q] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 Q_1(f_1^* f^* - f_1 f) c_r \sigma\, \mathrm{d}\Omega\, \mathrm{d}\boldsymbol{c}\, \mathrm{d}\boldsymbol{c}_1\,. \tag{3.26a}$$

The second symmetry may then be applied to this equation to give

$$\Delta[Q] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 Q_1^*(f_1 f - f_1^* f^*) c_r \sigma \, d\Omega \, d\boldsymbol{c} \, d\boldsymbol{c}_1, \tag{3.26b}$$

and a second application of the first symmetry gives

$$\Delta[Q] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 Q^*(f f_1 - f^* f_1^*) c_r \sigma \, d\Omega \, d\boldsymbol{c} \, d\boldsymbol{c}_1. \tag{3.26c}$$

Eqns (3.26), (3.26a), (3.26b), and (3.26c) may then be summed and the resultant equation divided by four, to give

$$\Delta[Q] = \tfrac{1}{4}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 (Q + Q_1 - Q^* - Q_1^*)(f f_1 - f^* f_1^*) c_r \sigma \, d\Omega \, d\boldsymbol{c} \, d\boldsymbol{c}_1 \tag{3.26d}$$

Also, eqn (3.26) may be written

$$\Delta[Q] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 Q f^* f_1^* c_r \sigma \, d\Omega \, d\boldsymbol{c}_1 \, d\boldsymbol{c} - \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 Q f f_1 c_r \sigma \, d\Omega \, d\boldsymbol{c}_1 \, d\boldsymbol{c},$$

and application of the second symmetry to the first term on the right-hand side converts $Q f^* f_1^*$ to $Q^* f f_1$. Therefore,

$$\Delta[Q] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 (Q^* - Q) f f_1 c_r \sigma \, d\Omega \, d\boldsymbol{c}_1 \, d\boldsymbol{c}. \tag{3.26e}$$

A similar transformation of eqn (3.26) yields

$$\Delta[Q] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 (Q_1^* - Q_1) f_1 f c_r \sigma \, d\Omega \, d\boldsymbol{c}_1 \, d\boldsymbol{c}. \tag{3.26f}$$

Finally, eqns (3.26e) and (3.26f) may be summed and the resulting equation halved, to give

$$\Delta[Q] = \tfrac{1}{2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 (Q^* + Q_1^* - Q - Q_1) f_1 f c_r \sigma \, d\Omega \, d\boldsymbol{c} \, d\boldsymbol{c}_1. \tag{3.26g}$$

The physical meaning of $\Delta[Q]$ is most readily apparent from eqn (3.26g) since $(Q^* + Q_1^* - Q - Q_1)$ represents the change in the quantity Q as the result of a collision of class $\boldsymbol{c}, \boldsymbol{c}_1 \rightarrow \boldsymbol{c}^*, \boldsymbol{c}_1^*$. This change is summed over all classes of collision and halved to allow for the double counting of collisions in the integration. That the integration is equivalent to summing over all collisions is readily seen if we note that

$$\tfrac{1}{2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{4\pi} n^2 f_1 f c_r \sigma \, d\Omega \, d\boldsymbol{c} \, d\boldsymbol{c}_1 = \tfrac{1}{2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \sigma_T c_r n^2 f_1 f \, d\boldsymbol{c}_1 \, d\boldsymbol{c} = \tfrac{1}{2} n^2 \overline{\sigma_T c_r},$$

which is in agreement with eqn (1.11) for the total number of collisions per unit time per unit volume of gas. The moment or transfer equation with the collision term in the form of eqn (3.26g) can therefore be derived independently of the Boltzmann equation and was, in fact, first derived by Maxwell (1867). The form of $\Delta[Q]$ in eqn (3.26d) is more closely allied to the Boltzmann formulation and states simply that the change in Q as a result of the inverse collisions is exactly equal to the change in Q as a result of the direct collisions.

If the quantity Q is either the mass m, momentum $m\boldsymbol{c}$, or energy $\frac{1}{2}mc^2$ of a molecule, the conservation of these quantities in collisions requires that $Q+Q_1-Q^*-Q_1^*=0$. Eqns. (3.26d) and (3.26g) then show that the collision integral $\Delta[Q]$ is zero, as would be expected from the physical meaning of the integral. The quantities m, $m\boldsymbol{c}$, and $\frac{1}{2}mc^2$ are called *collisional invariants*, while any that satisfy the condition $Q+Q_1-Q^*-Q_1^*=0$ are called *summational invariants*. It can be shown (e.g. Harris (1971) §4.2) that the collisional invariants, or linear combinations of them, are the only summational invariants. Therefore, if Q is a summational invariant, the collision integral $\Delta[Q]$ is 0 and Q can be written

$$Q = A\tfrac{1}{2}mc^2 + \boldsymbol{B}.m\boldsymbol{c} + C, \tag{3.28}$$

where A, $\boldsymbol{B}$, and C are constants.

The collision integral is zero in the three equations for the collisional invariants and the averages on the left-hand side of the equations can be expressed in terms of the macroscopic gas properties. The three equations are the *conservation equations* of gas dynamics. First, the equation for the *conservation of mass* is obtained by setting $Q = m$ in eqn (3.27), to give

$$\frac{\partial}{\partial t}(nm) + \nabla.(nm\overline{\boldsymbol{c}}) = 0 \tag{3.29}$$

or, using eqns (1.13) and (1.21),

$$\frac{\partial \rho}{\partial t} + \nabla.(\rho\,\boldsymbol{c}_0) = 0. \tag{3.30}$$

It is convenient to introduce the *substantial derivative*

$$\frac{\mathrm{D}}{\mathrm{D}t} \equiv \frac{\partial}{\partial t} + \boldsymbol{c}_0.\nabla \equiv \frac{\partial}{\partial t} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z} \tag{3.31}$$

to denote differentiation following the motion of the fluid element. The equation for the conservation of mass, or continuity equation, then becomes

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} = -\rho\,\nabla.\boldsymbol{c}_0. \tag{3.32}$$

Next, the equation for the *conservation of momentum* or *equation of motion* is obtained by setting $Q = m\boldsymbol{c}$ in eqn (3.27). This is the vector equation

$$\frac{\partial}{\partial t}(\rho\overline{\boldsymbol{c}}) + \nabla.(\rho\,\overline{\boldsymbol{cc}}) - \rho\boldsymbol{F} = 0. \tag{3.33}$$

But eqns (1.21) and (1.22) show that

$$\overline{\boldsymbol{cc}} = \overline{(\boldsymbol{c}' + \boldsymbol{c}_0)(\boldsymbol{c}' + \boldsymbol{c}_0)} = \overline{\boldsymbol{c}'\boldsymbol{c}'} + \boldsymbol{c}_0\boldsymbol{c}_0$$

and eqn (3.33) may be written

$$\rho \frac{\partial \boldsymbol{c}_0}{\partial t} + \boldsymbol{c}_0 \frac{\partial \rho}{\partial t} + \boldsymbol{c}_0 \nabla.(\rho \boldsymbol{c}_0) + \rho(\boldsymbol{c}_0.\nabla)\boldsymbol{c}_0 + \nabla.(\rho \overline{\boldsymbol{c}'\boldsymbol{c}'}) - \rho \boldsymbol{F} = 0.$$

The continuity equation (3.30) enables the second and third terms to be removed, eqn (3.31) enables the first and fourth terms to be combined as a substantial derivative, and eqn (1.20) enables the fifth term to be written in terms of the pressure and the viscous stress tensor, i.e.

$$\rho \frac{\mathrm{D}\boldsymbol{c}_0}{\mathrm{D}t} = -\nabla p + \nabla.\boldsymbol{\tau} + \rho \boldsymbol{F}. \tag{3.34}$$

The vector and tensor quantities are clarified if we also write the momentum equation in the x-direction in Cartesian coordinates. This is

$$\rho \frac{\mathrm{D}u_0}{\mathrm{D}t} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + \rho F_x.$$

Finally, Q may be set equal to $\frac{1}{2}mc^2$ in eqn (3.27) to give the equation for the *conservation of energy* as

$$\frac{\partial}{\partial t}(\tfrac{1}{2}\rho \overline{c^2}) + \nabla.(\tfrac{1}{2}\rho \overline{\boldsymbol{c}c^2}) - \rho \boldsymbol{c}_0.\boldsymbol{F} = 0. \tag{3.35}$$

The averages over the molecular velocities may again be converted to averages over the peculiar and stream velocities, to give

$$\overline{c^2} = \overline{c'^2} + c_0^2$$

and

$$\overline{\boldsymbol{c}c^2} = \overline{\boldsymbol{c}'c'^2} + \boldsymbol{c}_0(\overline{c'^2} + c_0^2) + 2u_0\overline{\boldsymbol{c}'u'} + 2v_0\overline{\boldsymbol{c}'v'} + 2w_0\overline{\boldsymbol{c}'w'}.$$

The macroscopic quantities may then be introduced and the final result is

$$\rho \frac{\mathrm{D}e}{\mathrm{D}t} = -p\nabla.\boldsymbol{c}_0 + \Phi - \nabla.\boldsymbol{q}. \tag{3.36}$$

The quantity Φ is called the *dissipation function* and may be written in Cartesian coordinates as

$$\begin{aligned}\Phi = {} & \tau_{xx}\frac{\partial u}{\partial x} + \tau_{yy}\frac{\partial v}{\partial y} + \tau_{zz}\frac{\partial w}{\partial z} + \tau_{xy}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) \\ & + \tau_{yz}\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right) + \tau_{zx}\left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}\right).\end{aligned} \tag{3.37}$$

Since the momentum equation (3.34) is a vector equation, the equations for the conservation of mass, momentum, and energy constitute five equations. The dependent variables include the three velocity components and, if the

equation of state is taken into account, two of the three thermodynamic properties p, ρ, and T. The viscous stress tensor τ contains nine components but, because of its symmetry and the relationship of the sum of the diagonal components to the pressure, it contributes only five dependent variables. Finally, the heat flux vector $\boldsymbol{q}$ contributes a further three dependent variables. The conservation equations therefore contain a total of thirteen dependent variables and do not form a determinate set. The *inviscid* case with τ and $\boldsymbol{q}$ both equal to zero does result in a determinate set, and these are called the *Euler equations*.

The application of moment equations other than the conservation equations generally requires the evaluation of the collision integral $\Delta[Q]$. This process may be illustrated by the special case of $Q = u^2$ for Maxwell molecules. The substitution of this value of Q into eqn (3.26e), together with eqn (2.40) for the cross-section of Maxwell molecules, yields

$$\Delta[u^2] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{2\pi}\int_{0}^{\infty} n^2(u^{*2} - u^2) f f_1 W_0 \left(\frac{2\kappa}{m}\right)^{\frac{1}{2}} \mathrm{d}W_0 \,\mathrm{d}\varepsilon \,\mathrm{d}\boldsymbol{c} \,\mathrm{d}\boldsymbol{c}' .$$

The advantage of Maxwell molecules is that the molecular velocities appear only in the term $u^{*2} - u^2$, which may be written as $(u^* - u)^2 + 2u(u^* - u)$. Then with u and u_1 identified with u_2 and u_1 in eqns (2.4) and (2.5), and using eqn (2.22),

$$u^* - u = \tfrac{1}{2}(u_r - u_r^*) = \tfrac{1}{2}\{(1 - \cos\chi)u_r - \sin\chi \sin\varepsilon\,(v_r^2 + w_r^2)^{\frac{1}{2}}\} .$$

Therefore,

$$u^{*2} - u^2 = \tfrac{1}{4}\{(1 - \cos\chi)^2 u_r^2 - 2(1 - \cos\chi)u_r \sin\chi \sin\varepsilon\,(v_r^2 + w_r^2)^{\frac{1}{2}} + \sin^2\chi \sin^2\varepsilon (v_r^2 + w_r^2)\} + u\{(1 - \cos\chi)u_r - \sin\chi \sin\varepsilon (v_r^2 + w_r^2)^{\frac{1}{2}}\} .$$

The integration over ε may be carried out first, to give,

$$\int_0^{2\pi} (u^{*2} - u^2)\,\mathrm{d}\varepsilon = \pi(u_r^2 + 2uu_r)(1 - \cos\chi) - \tfrac{1}{4}\pi(3u_r^2 - c_r^2)\sin^2\chi .$$

Eqn (2.3) shows that $u_r^2 + 2uu_r = u_1^2 - u^2$ and, since u and u_1 are described by the same distribution, this term disappears in the integration over velocity space. The remaining term contains only relative velocities, and the result is independent of whether these are expressed as differences between velocities or peculiar velocities and, therefore, $\Delta[u^2] = \Delta[u'^2]$. Eqns (2.3) and (1.22) give $u_r^2 = u'_1 - 2u'_1 u' + u'^2$ and the double integration over velocity space yields $2\overline{u'^2}$. A similar result holds for c_r^2, and the collision integral becomes

$$\Delta[u^2] = -\frac{3\pi}{2}\left(\frac{2\kappa}{m}\right)^{\frac{1}{2}} n^2 (\overline{u'^2} - \overline{c'^2}/3) \int_0^{\infty} \sin^2\chi \, W_0 \,\mathrm{d}W_0 .$$

Finally, from eqns (1.25) and (1.26),

$$\Delta[u^2] = -\frac{3\pi}{2} A_2(5) \left(\frac{2\kappa}{m}\right)^{1/2} \frac{n}{m} \tau_{xx}, \tag{3.38}$$

where

$$A_2(5) = \int_0^\infty \sin^2\chi \, W_0 \, dW_0$$

for Maxwell molecules. $A_2(5)$ is a pure number and its value may be obtained through eqn (2.39) as 0.436.

The collision integral $\Delta[u^2]$ is also readily evaluated for the special case of the VHS model in which the diameter is inversely proportional to the square root of the relative speed, or $\upsilon = \frac{1}{2}$. As with the Maxwell gas, all molecules in this gas have the same collision probability. Eqn (2.30) for the differential cross-section of hard sphere molecules may be combined with eqn (2.34) and substituted into eqn (3.26e). Then, if it is noted that $d\Omega = \sin\chi \, d\chi \, d\varepsilon$, we have

$$\Delta[u^2] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_0^{2\pi}\int_0^{\pi} n^2 (u^{*2} - u^2) f f_1 (c_{r,ref} d_{ref}^2/4) \sin\chi \, d\chi \, d\varepsilon \, dc \, dc' .$$

The integration over ε may be carried out in exactly the same manner as in the Maxwell gas case, so that

$$\Delta[u^2] = -\frac{3\pi}{2} n^2 (\overline{u'^2} - \overline{c'^2}/3) \frac{c_{r,ref} d_{ref}^2}{4} \int_0^\pi \sin^3\chi \, d\chi ,$$

and the final result is

$$\Delta[u^2] = \tfrac{1}{2} c_{r,ref} d_{ref}^2 (n/m) \tau_{xx} . \tag{3.39}$$

3.4 The *H*-theorem and equilibrium

Consider a spatially homogeneous volume of a simple dilute monatomic gas that is free of any external force. The Boltzmann equation (3.20) may be simplified for such a gas since the number density n is a constant, spatial derivatives $\partial/\partial r$ are zero, and the external force F is zero. The equation becomes

$$\frac{\partial f}{\partial t} = n \int_{-\infty}^{\infty}\int_0^{4\pi} (f^* f_1^* - f f_1) c_r \sigma \, d\Omega \, dc_1 . \tag{3.40}$$

Over a small time interval, f changes to $f + \Delta f$ and the fractional change is $\Delta f/f$ or $\Delta(\ln f)$. Boltzmann's H-function is the mean value of $\ln(nf)$, i.e.

$$H = \overline{\ln(nf)}$$

or, using eqn(3.3),

$$H = \int_{-\infty}^{\infty} f \ln(nf)\, \mathrm{d}\boldsymbol{c}. \tag{3.41}$$

The quantity Q in the moment equation (3.27) may then be set equal to $\ln(nf)$ and, with the collision integral $\Delta[Q]$ in the form given by eqn (3.26d), we have

$$\frac{\partial H}{\partial t} = \frac{n}{4} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{0}^{4\pi} (\ln f + \ln f_1 - \ln f^* - \ln f_1^*)(f^* f_1^* - f f_1)\, c_r\, \sigma\, \mathrm{d}\Omega\, \mathrm{d}\boldsymbol{c}\, \mathrm{d}\boldsymbol{c}_1$$

$$= \frac{n}{4} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{0}^{4\pi} \ln(f f_1 / f^* f_1^*)(f^* f_1^* - f f_1)\, c_r\, \sigma\, \mathrm{d}\Omega\, \mathrm{d}\boldsymbol{c}\, \mathrm{d}\boldsymbol{c}_1. \tag{3.42}$$

If $\ln(ff_1/f^*f_1^*)$ is positive, then $(f^*f_1^* - ff_1)$ must be negative and vice versa. The integral on the right hand side of eqn (3.42) is, therefore, either negative or zero and H can never increase, i.e.

$$\frac{\partial H}{\partial t} < 0. \tag{3.43}$$

This result is known as Boltzmann's H-theorem.

The question which now arises is whether H decreases without limit to $-\infty$, or tends to a finite value and thereafter remains constant. Now, as $c \to \infty$, $f \to 0$ and $\ln f \to -\infty$, and it appears that the integral for H in eqn (3.41) may diverge. However, since the energy of the gas is finite, the integral

$$\int_{-\infty}^{\infty} f c^2\, \mathrm{d}\boldsymbol{c}$$

must converge. Therefore, for H to diverge, the approach of f to 0 would be more rapid than that of $\exp(-c^2)$ to 0 and, since $\exp(-x^2)x^n \to 0$ as $x \to \infty$ for all values of n, H must converge. Therefore, for any initial distribution of molecules in velocity space, the distribution will alter with time in such a way that H decreases monotonically to a finite lower bound. At subsequent times,

$$\frac{\partial H}{\partial t} = 0$$

and eqn (3.42) shows that this requires

$$f^* f_1^* - f f_1 = 0 \tag{3.44}$$

or, equivalently,

$$\ln f + \ln f_1 = \ln f^* + \ln f_1^*. \tag{3.45}$$

A comparison of eqns (3.45) and (3.40) shows that the stationary state for H is also a stationary state for f. We therefore have an *equilibrium state* in which the probable number of molecules in any element of velocity space remains constant with time.

Eqn (3.45) shows that, in the equilibrium state, $\ln f$ is a summational invariant. Therefore, from eqn (3.28), the necessary and sufficient condition for equilibrium is that

$$\ln f = A\tfrac{1}{2}mc^2 + \boldsymbol{B}.m\boldsymbol{c} + C. \tag{3.46}$$

This equation may be written in terms of the thermal velocity components as

$$\ln f = A\tfrac{1}{2}mc'^2 + m(A\boldsymbol{c}_0 + \boldsymbol{B}).\boldsymbol{c}' + A\tfrac{1}{2}mc_0^2 + \boldsymbol{B}.m\boldsymbol{c}_0 + C.$$

Since there can be no preferred direction in the equilibrium gas, the distribution must be isotropic. This requires that the coefficient of $\boldsymbol{c}'$ be zero, or

$$\boldsymbol{B} = -A\boldsymbol{c}_0.$$

Therefore,

$$\ln f = \tfrac{1}{2}Amc'^2 - \tfrac{1}{2}Amc_0^2 + C,$$

or,

$$f = \exp(\tfrac{1}{2}Amc'^2 - \tfrac{1}{2}Amc_0^2 + C).$$

Since f is bounded, the coefficient of c'^2 must be negative and, for convenience, we introduce a new constant by setting $\tfrac{1}{2}Am = -\beta^2$. We therefore have

$$f = \exp(C + \beta^2 c_0^2)\exp(-\beta^2 c'^2).$$

The constant C may be eliminated through the normalization condition expressed in eqn (3.2). This requires that

$$\int_{-\infty}^{\infty} f\,\mathrm{d}\boldsymbol{c} = \exp(C + \beta^2 c_0^2)\int_{-\infty}^{\infty}\exp(-\beta^2 c'^2)\,\mathrm{d}\boldsymbol{c}' = 1.$$

But,

$$\int_{-\infty}^{\infty}\exp(-\beta^2 c'^2)\,\mathrm{d}\boldsymbol{c}' = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\exp\{-\beta^2(u'^2 + v'^2 + w'^2)\}\,\mathrm{d}u'\,\mathrm{d}v'\,\mathrm{d}w'$$

$$= \int_{-\infty}^{\infty}\exp(-\beta^2 u'^2)\,\mathrm{d}u'\int_{-\infty}^{\infty}\exp(-\beta^2 v'^2)\,\mathrm{d}v'\int_{-\infty}^{\infty}\exp(-\beta^2 w'^2)\,\mathrm{d}w'.$$

The list of standard integrals in Appendix B shows that each component integral is $\pi^{\frac{1}{2}}/\beta$, so that

$$\exp(C + \beta^2 c_0^{\,2}) = \beta^3/\pi^{3/2}.$$

Therefore, the final result for the *equilibrium* or *Maxwellian distribution function* f_0 is

$$f_0 = (\beta^3/\pi^{3/2})\exp(-\beta^2 c'^2). \tag{3.47}$$

The constant β may be related to the temperature of the gas. Eqns (1.29), (3.4), and (3.47) show that

$$3RT/2 = \tfrac{1}{2}\overline{c'^2} = \frac{\beta^3}{2\pi^{3/2}} \int_{-\infty}^{\infty} c'^2 \exp(-\beta^2 c'^2)\, d\mathbf{c}$$

or again using the standard integrals of Appendix B,

$$\beta^2 = (2RT)^{-1} = m/(2kT). \tag{3.48}$$

Since we are dealing with an equilibrium gas, the translational temperature T_{tr} calculated through eqn (1.29) is equal to the thermodynamic temperature T and the subscript has been dropped. Substitution of eqn (3.48) into (3.47) gives

$$f_0 = \{m/(2\pi kT)\}^{3/2} \exp\{-mc'^2/(2kT)\} \tag{3.47a}$$

as an alternative definition of f_0.

In the case of a gas mixture in equilibrium, it can be shown (Chapman and Cowling 1952, p. 84) that the distribution function of species p follows directly as

$$f_{0,p} = \{m_p/(2\pi kT)\}^{3/2} \exp\{-m_p c'^2_p/(2kT)\}. \tag{3.49}$$

The Maxwellian distribution applies also to the translational velocities of diatomic and polyatomic molecules. This result has been proven for the special case of rough-sphere molecules (Chapman and Cowling 1970, §11.4) using an extension of the method that has been applied here to monatomic molecules. The collision mechanics of more realistic models are sufficiently complex to rule out this approach. However, if it is assumed that, for each collision, a collision with the velocity components reversed is equally likely, the generalized Boltzmann equation leads to a derivation that is applicable to all molecular models (Chapman and Cowling 1970, §11.3). Moreover, for equilibrium situations, the methods of statistical mechanics are available and provide proofs that are more general than those from kinetic theory.

3.5 The Chapman–Enskog theory

The Chapman–Enskog method provides a solution of the Boltzmann equation for a restricted set of problems in which the distribution function f is perturbed by a small amount from the equilibrium Maxwellian form. It is assumed that the distribution function may be expressed in the form of the power series

$$f = f^{(0)} + \varepsilon_0 f^{(1)} + \varepsilon_0^2 f^{(2)} + \dots, \tag{3.50}$$

where ε_0 is a parameter which may be regarded as a measure of either the mean collision time or the Knudsen number. The first term $f^{(0)}$ is the Maxwellian distribution f_0 for an equilibrium gas, and an alternative form of the expression is

$$f = f_0(1 + \Phi_1 + \Phi_2 + \dots). \tag{3.51}$$

The equilibrium distribution function constitutes the known first-order solution of this equation and the second-order solution requires the determination of the parameter Φ_1.

Solutions of the Boltzmann equation for

$$f = f_0(1 + \Phi_1) \tag{3.52}$$

were obtained independently by Enskog and Chapman, and these form the subject matter of the classical work by Chapman and Cowling (1952). For a simple gas, Φ_1 depends only on the density, stream velocity, and temperature of the gas so that the resulting solution constitutes a *normal solution* of the Boltzmann equation. Since f_0 satisfies the equations

$$\int_{-\infty}^{\infty} f \, \mathrm{d}\boldsymbol{c} = 1,$$

$$\int_{-\infty}^{\infty} \boldsymbol{c} f \, \mathrm{d}\boldsymbol{c} = \boldsymbol{c}_0,$$

and

$$\int_{-\infty}^{\infty} c'^2 f \, \mathrm{d}\boldsymbol{c} = 3RT,$$

Φ_1 must be such that

$$\int_{-\infty}^{\infty} \Phi_1 f_0 \, \mathrm{d}\boldsymbol{c} = 0,$$

$$\int_{-\infty}^{\infty} \boldsymbol{c} \Phi_1 f_0 \, \mathrm{d}\boldsymbol{c} = 0,$$

and

$$\int_{-\infty}^{\infty} c'^2 \Phi_1 f_0 \, \mathrm{d}\boldsymbol{c} = 0. \tag{3.53}$$

It may further be shown that Φ_1 must have the form

$$\Phi_1 = -\frac{1}{n} A \boldsymbol{c}' . \frac{\partial}{\partial \boldsymbol{r}} (\ln T) + B \boldsymbol{c}'^0 \boldsymbol{c}' : \frac{\partial \boldsymbol{c}_0}{\partial \boldsymbol{r}} \tag{3.54}$$

where A and B are functions of T and c'. The superscript ^{0} above a tensor indicates that the sum of the diagonal components is zero. In the component or subscript notation of Chapter 1,

$$c'^0_i c'_j = c'_i c'_j - c'^2 \delta_{ij}/3 .$$

The double product of two tensors is a scalar quantity which may be written in the subscript notation as

$$c'^0_i c'_j \frac{\partial c_{0i}}{\partial x_j},$$

or in full Cartesian component form as

$$(u'^2 - c'^2/3)\frac{\partial u_0}{\partial x} + u'v'\frac{\partial u_0}{\partial y} + u'w'\frac{\partial v_0}{\partial z} + v'u'\frac{\partial v_0}{\partial x} + (v'^2 - c'^2/3)\frac{\partial v_0}{\partial y}$$

$$+ v'w'\frac{\partial v_0}{\partial z} + w'u'\frac{\partial w_0}{\partial x} + w'v'\frac{\partial w_0}{\partial y} + (w'^2 - c'^2/3)\frac{\partial w_0}{\partial z},$$

where $c'^2 = u'^2 + v'^2 + w'^2$.

Eqns (3.52) and (3.54) may be combined with eqns (1.26) and (1.33) to give expressions for the shear stress tensor and the heat flux vector as linear functions of the velocity and temperature gradients, respectively. The coefficients of the gradients may be identified with the familiar coefficients of viscosity and heat conduction. The coefficients A and B of eqn (3.54) are usually obtained as series of Sonine polynomials. Since the word 'order' has already been applied to the specification of the number of terms in eqn (3.51), the solution involving n terms of the Sonine polynomial series is usually referred to as the 'nth approximation'. The molecular model enters the analysis and it can be shown (Vincenti and Kruger 1965, Chap. X, §7) that the first approximation to the coefficient of viscosity μ for a monatomic gas is

$$\mu = \frac{(5/8)(\pi m k T)^{1/2}}{\{m/(4kT)\}^4 \int_0^\infty c_r^7 \sigma_\mu \exp\{-m c_r^2/(4kT)\} \mathrm{d}c_r} . \tag{3.55}$$

The first approximation to the coefficient of heat conduction K is related to the coefficient of viscosity by

$$K = (15/4)(k/m)\mu = (15/4)R\mu, \tag{3.56}$$

so that the Prandtl number in a monatomic gas, for which the specific heat at constant pressure c_p is equal to $5R/2$, is

$$(Pr) = \mu c_p / K = 2/3. \tag{3.57}$$

The viscosity cross-section σ_μ for a hard sphere gas is given by eqn (2.32). Eqn (3.55) is then readily evaluated to give the coefficient of viscosity in a hard sphere gas as

$$\mu = (5/16)(\pi m k T)^{1/2}/\sigma_T = (5/16)(RT/\pi)^{1/2}(m/d^2). \tag{3.58}$$

The diameter of a hard sphere gas that has a coefficient of viscosity μ_{ref} at temperature T_{ref} is therefore

$$d = \{(5/16)(mkT_{ref}/\pi)^{1/2}/\mu_{ref}\}^{1/2} \tag{3.59}$$

and is, of course, independent of the relative velocity and relative translational energy.

For VHS molecules, the cross-section is inversely proportional to a power of the relative speed. The definition can be extended to include the total cross-section and the relative translational energy, i.e.

$$\sigma_T/\sigma_{T,ref} = (d/d_{ref})^2 = (c_r/c_{r,ref})^{-2\upsilon} = (E_t/E_{t,ref})^{-\upsilon}, \tag{3.60}$$

where υ is a constant and $\sigma_{T,ref}$ and d_{ref} are the total cross-section and diameter that apply when the relative speed is $c_{r,ref}$. As noted in §2.6, the viscosity cross-section is still given by eqn (2.32), and eqn (3.55) becomes

$$\mu = \frac{(15/8)(\pi m k)^{1/2}(4k/m)^{\upsilon} T^{1/2+\upsilon}}{\Gamma(4-\upsilon)\,\sigma_{T,ref}\, c_{r,ref}^{2\upsilon}}. \tag{3.61}$$

The VHS cross-section defined by eqn (3.60) therefore leads to a coefficient of viscosity proportional to a fixed power of the temperature.

The evaluation of eqn (3.55) for the inverse power law model of §2.4 gives

$$\mu = \frac{5m(RT/\pi)^{1/2}(2mRT/\kappa)^{2/(\eta-1)}}{8A_2(\eta)\Gamma\{4-2/(\eta-1)\}}. \tag{3.62}$$

$A_2(\eta)$ is a numerical factor defined by

$$A_2(\eta) \equiv \int_0^\infty \sin^2\chi\, W_0\, \mathrm{d}W_0$$

and for which Chapman and Cowling (1952) give the following values

η	5	7	9	11	15	21	∞
$A_2(\eta)$	0.436	0.357	0.332	0.319	0.309	0.307	0.333

While the coefficients of viscosity and heat conduction can be regarded as continuum gas properties, it will be found that an essential feature of a successful molecular model for rarefied gas flow studies is that it should reproduce the viscosity coefficient of the real gas, and also the temperature dependence of this coefficient. This is essentially an empirical observation, but some analytical support may be obtained from a comparison of the collision integral $\Delta[u^2]$ for Maxwell molecules (eqn (3.38)) with the same collision integral for VHS molecules with $\upsilon = \frac{1}{2}$ (eqn (3.39)), when both of

these are expressed in terms of the coefficient of viscosity. The coefficient of viscosity in a Maxwell gas is obtained by setting $\eta = 5$ in eqn (3.62), to give

$$\mu = \frac{2kT}{3\pi A_2(5)}\left(\frac{m}{2\kappa}\right)^{1/2}. \tag{3.63}$$

The coefficient of viscosity in a VHS gas with $\upsilon = 1/2$ is, from eqn (3.61)

$$\mu = \frac{2kT}{\sigma_{T,\mathrm{ref}} c_{r,\mathrm{ref}}}. \tag{3.64}$$

The substitution of eqn (3.63) into eqn (3.38) and the substitution of eqn (3.64) into eqn (3.39) both lead to exactly the same result for the collision integral. This is

$$\Delta[u^2] = \frac{p}{m}\frac{\tau_{xx}}{\mu}. \tag{3.65}$$

Therefore, as long as the viscosity coefficients are matched, the collision integrals are identical. The matching of the viscosity coefficient means that the two models give the same result in the continuum limit. The equality of the collision integrals mean that they also give the same result in approximate methods, such as the Mott–Smith theory (§8.2) for shock wave structure, that employ this collision integral and apply over the full range of Knudsen number.

The inverse power law model and the VHS model both lead to a power law temperature dependence of the coefficient of viscosity. That is,

$$\mu \propto T^{\omega}. \tag{3.66}$$

where

$$\omega = 1/2 + \upsilon = 1/2(\eta + 3)/(\eta - 1). \tag{3.67}$$

For a gas that has a coefficient of viscosity proportional to T^{ω} and with the value μ_{ref} at temperature T_{ref}, eqns (3.60), (3.61), and (3.67) show that the molecular diameter is related to the relative translational energy in the collision by

$$d = \left(\frac{(15/8)(m/\pi)^{1/2}(kT_{\mathrm{ref}})^{\omega}}{\Gamma(9/2-\omega)\,\mu_{\mathrm{ref}} E_t^{\omega-1/2}}\right)^{1/2}. \tag{3.68}$$

Alternatively, the formulation could have been for the total cross-section as a function of either the magnitude of the relative velocity or the relative translational energy.

In the case of a gas mixture, the diameter may be calculated for each of the molecules in the collision and the total cross-section can be calculated from mean diameter, as was done for eqn (2.31). However, it is desirable to define separate values of σ_T and ω for the collisions between unlike molecules and, for this purpose, the values obtained from the diffusion coefficient may be more appropriate than those obtained from the viscosity coefficient.

The Chapman–Enskog first approximation to the diffusion coefficient in a binary gas mixture of species 1 and species 2 molecules is

$$D_{12} = \frac{(3/16)(2\pi kT/m_r)^{1/2}}{\{m_r/(2kT)\}^3 \int_0^\infty c_r^5 \sigma_M \exp\{-m_r c_r^2/(2kT)\} dc_r}. \tag{3.69}$$

The momentum cross-section was defined in eqn (2.29) and, for both the hard sphere and VHS models, it was shown to be equal to the total cross-section. This allows the expression for the diffusion coefficient to be evaluated for the VHS model in terms of the reference cross-section $(\sigma_{T,ref})_{12}$ at the temperature T_{ref}, to give

$$D_{12} = \frac{(3/8)\pi^{1/2}(2kT/m_r)^{\omega_{12}}}{\Gamma(7/2-\omega_{12}) n (\sigma_{T,ref})_{12} c_{r,ref}^{2(\omega_{12}-1/2)}}. \tag{3.70}$$

For the special case of a hard sphere gas, this simplifies to

$$D_{12} = \frac{3(2\pi kT/m_r)^{1/2}}{16 n (\sigma_T)_{12}}. \tag{3.71}$$

If the cross-collisions in a binary gas mixture are specified by the value $(D_{12})_{ref}$ of the diffusion coefficient at the reference temperature T_{ref}, the collision diameter is related to the relative translational energy E_t in the collision by

$$d_{12} = \left(\frac{(3/8)(2kT_{ref})^{\omega_{12}}}{\Gamma(7/2-\omega_{12})(\pi m_r)^{1/2} n (D_{12})_{ref} (2E_t)^{\omega_{12}-1/2}} \right)^{1/2}. \tag{3.72}$$

A typical difference between this diameter and that defined by substituting the viscosity-based diameters of the individual species into eqn (2.31) is shown in Fig. 3.3. This is for a mixture of argon and helium and it is based on the values of the transport properties that are quoted in Chapman and Cowling (1970). If the hard sphere model is used in place of the VHS model, the constant diameters for argon and helium atoms are 3.66×10^{-10} m and 2.19×10^{-10} m, respectively. The effective diameter of the argon–helium pair from eqn (2.31) is 2.93×10^{-10} m, while the diffusion-based diameter is $2.62\times^{-10}$ m. Part of the discrepancy may be due to the fact that the viscosity and diffusion coefficients are the first approximation to the exact analytical values, and the correction that is introduced by the second approximation is much higher for the diffusion coefficient than for the viscosity coefficient. However, it is almost certain that the difference between the viscosity- and diffusion-based diameters is largely due to the incorrect ratio of the momentum cross-section to the diffusion cross section. The VSS model was introduced in order to correct this ratio and the preceding VHS results may be reformulated for the VSS model.

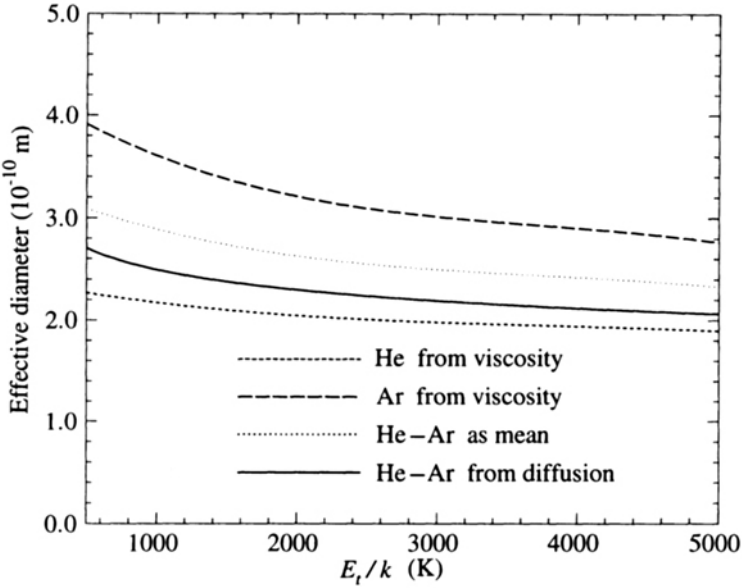


Fig. 3.3 Effective molecular diameters based on the VHS model for an argon–helium mixture.

When the variable soft sphere (or VSS) viscosity cross-section of eqn (2.37) is used in place of the VHS value in eqn (3.55), the coefficient of viscosity in a VSS gas becomes

$$\mu = \frac{5(\alpha+1)(\alpha+2)(\pi m k)^{1/2}(4k/m)^{\upsilon} T^{1/2+\upsilon}}{16\alpha\Gamma(4-\upsilon)\,\sigma_{T,\mathrm{ref}} c_{r,\mathrm{ref}}^{2\upsilon}}. \tag{3.73}$$

The viscosity-based diameter of a VSS molecule is, therefore,

$$d = \left(\frac{5(\alpha+1)(\alpha+2)(m/\pi)^{1/2}(kT_{\mathrm{ref}})^{\omega}}{16\alpha\Gamma(9/2-\omega)\,\mu_{\mathrm{ref}} E_t^{\omega-1/2}}\right)^{1/2}. \tag{3.74}$$

A similar procedure gives the diffusion coefficient in a VSS gas as

$$D_{12} = \frac{3(\alpha_{12}+1)\pi^{1/2}(2kT/m_r)^{\omega_{12}}}{16\Gamma(7/2-\omega_{12})n(\sigma_{T,\mathrm{ref}})_{12} c_{r,\mathrm{ref}}^{2(\omega_{12}-1/2)}}, \tag{3.75}$$

and the diffusion-based diameter becomes

$$d_{12} = \left(\frac{3(\alpha_{12}+1)(2kT_{\mathrm{ref}})^{\omega_{12}}}{16\Gamma(7/2-\omega_{12})(\pi m_r)^{1/2} n(D_{12})_{\mathrm{ref}}(2E_t)^{\omega_{12}-1/2}}\right)^{1/2}. \tag{3.76}$$

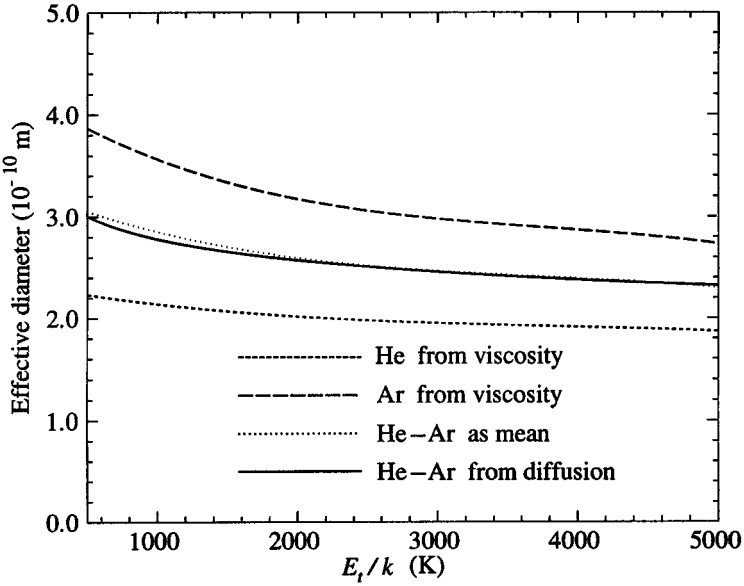


Fig. 3.4 Effective molecular diameters based on the VSS model for an argon–helium mixture.

Koura and Matsumoto (1992) have presented values of α for common gases, and their values for helium and argon are used in Fig. 3.4 to show the resulting VSS molecular diameters. The effective diameters based on the viscosity and diffusion coefficients are now consistent, and the VSS model is clearly superior to the VHS model.

Both the VHS and VSS models employ a power law relationship between the cross-section or diameter and the relative speed or collision energy. Alternative choices for the variation of cross-section with relative speed are available. A particularly interesting case due to Kuščer (private communication) is to set

$$\sigma_T = \sigma_{T,\infty}\{1 + 6kT_s/E_t\}, \tag{3.77}$$

where T_s is a reference temperature and $\sigma_{T,\infty}$ is the total cross-section in the limiting case of infinite c_r. The evaluation of eqn (3.55) for this cross-section gives

$$\mu = \frac{5}{16}\frac{(\pi mkT)^{1/2}}{\sigma_{T,\infty}}\frac{T}{T+T_s}. \tag{3.78}$$

This temperature dependence of the viscosity coefficient is identical to that in Sutherland's formula which reproduces (Chapman and Cowling 1970) the experimental data for many real gases over a considerable range of temperature. The Sutherland model is the combination of a hard sphere of diameter σ_∞ with a power law of attraction. The replacement of the hard

sphere component by the VHS model of eqn (3.60) leads to a VHS model that includes both attractive and repulsive power laws. The cross-section for this model is, therefore,

$$\sigma_T = \sigma_{T,ref}\left(\frac{E_{t,ref}}{E_t}\right)^{\omega-\frac{1}{2}}\left(1+\frac{6kT_s}{E_t}\right), \tag{3.79}$$

However, while this model has two terms with separate exponents for E_t, the values of the exponents are restricted to $-(\omega-\frac{1}{2})$ and $-(\omega+\frac{1}{2})$. This combination cannot reproduce the behaviour of, for example, the Lennard–Jones model. It is preferable to employ the 'generalized VHS' (or GHS) model (Hassan and Hash, 1993) which has a cross-section defined by

$$\frac{\sigma_T}{s_0^2} = \alpha_1\left(\frac{E_t}{\varepsilon}\right)^{-\psi_1} + \alpha_2\left(\frac{E_t}{\varepsilon}\right)^{-\psi_2}. \tag{3.80}$$

Here, s_0 is the radius at which the potential is zero and ε is related to the maximum attractive force (or 'well-depth'). These quantities may be visualized by reference to Fig. 1.3. ψ_1 and ψ_2 are, of course, the inverse power laws for the variation of the cross-section with the relative energy. These all have physical significance, while α_1 and α_2 are dimensionless numbers that are used to better fit the expression to the experimental database. While six parameters have been introduced, these could be reduced algebraically to four, and eqn (3.80) is essentially a four parameter model for the variation of collision cross-section with relative velocity. By contrast, the hard sphere model has zero parameters and the VHS model has one. Eqn (3.55) for the coefficient of viscosity may be evaluated for this model, to give

$$\mu = \frac{(5/8)(\pi m k T)^{\frac{1}{2}}/s_0^2}{(\alpha_1/3)\Gamma(4-\psi_1)(kT/\varepsilon)^{-\psi_1} + (\alpha_2/3)\Gamma(4-\psi_2)(kT/\varepsilon)^{-\psi_2}}. \tag{3.81}$$

It is not possible to define a reference cross-section or diameter for this model and, for it to be applied to a particular gas, numerical values must be supplied for all of the parameters in eqn (3.80). The existing database in the literature for the transport properties can be used for this purpose if the constants ψ_1 and ψ_2 in eqn (3.80) can be related to the power laws in a widely used model such as the Lennard–Jones model. The force between the molecules in the Lennard–Jones model is

$$F = \frac{\kappa}{r^{\eta}} - \frac{\kappa'}{r^{\eta'}}, \tag{3.82}$$

but the viscosity coefficient cannot be written in a closed form. However, it has been shown by Chapman and Cowling (1970, §10.41) that, if the power law of attraction is weak in comparison with that of repulsion (i.e. that η' is much less than η), an approximate expression for the coefficient of viscosity is

$$\mu = \mu_0\left(1 + S/T^{(\eta-\eta')(\eta-1)}\right)^{-1}. \tag{3.83}$$

Here, μ_0 is the coefficient of viscosity without the attractive component, and S is a function of the constants in the model. Eqn (3.81) can be written

$$\mu = \frac{15(\pi m k T)^{\frac{1}{2}}(kT/\varepsilon)^{\psi_1}}{8\alpha_1\Gamma(4-\psi_1)s_0^2}\left(1+\frac{\alpha_2\Gamma(4-\psi_2)}{\alpha_1\Gamma(4-\psi_1)}(kT/\varepsilon)^{\psi_1-\psi_2}\right)^{-1}.$$

The power of T in the first part of this equation may be equated with that in eqn (3.66) and the power of T in the second part may be similarly equated with that in eqn (3.83). This gives two equations for ψ_1 and ψ_2 as functions of η and η' and these may be solved to give

$$\psi_1 = 2/(\eta-1)$$

and (3.84)

$$\psi_2 = (2+\eta-\eta')/(\eta-1).$$

Therefore, for the Lennard–Jones 6-12 potential with η=13 and η'=7, the power laws in the expression for the cross-section are ψ_1=1/6 and ψ_2=2/3. The parameters for a cross-collision in a gas mixture may be fitted to the diffusion coefficient. For the GHS model, this may be written

$$D_{12} = \frac{(3/8)(2\pi kT/m_r)^{\frac{1}{2}}/(s_0^2 n)}{\alpha_1\Gamma(3-\psi_1)(kT/\varepsilon)^{-\psi_1}+\alpha_2\Gamma(3-\psi_2)(kT/\varepsilon)^{-\psi_2}}. \qquad (3.85)$$

The Chapman–Enskog distribution function is, itself, directly applicable to many problems in rarefied gas dynamics. The evaluation of the constants A and B in eqn (3.54) allows eqn (3.52) to be written

$$f = f_0\left(1-\frac{4K\beta^2}{5nk}(\beta^2 c'^2-5/2)\,\boldsymbol{c}'.\frac{\partial(\ln T)}{\partial \boldsymbol{r}}-\frac{4\mu\beta^4}{\rho}\boldsymbol{c}'^0\boldsymbol{c}':\frac{\partial \boldsymbol{c}_0}{\partial \boldsymbol{r}}\right). \qquad (3.86)$$

The physical implications of this equation are more readily appreciated if we consider the special case with the stream velocity in the x-direction and in which gradients exist only in the y-direction. Eqn (3.86) becomes

$$f = f_0\left(1-\frac{4K\beta^2 v'}{5nk}(\beta^2 c'^2-5/2)\frac{\partial(\ln T)}{\partial y}-\frac{4\mu s\beta^3 u'v'}{\rho}\frac{\partial(\ln u_0)}{\partial y}\right). \qquad (3.87)$$

The Chapman–Enskog theory requires the two perturbation terms to be small compared with unity. This condition will be violated for sufficiently large thermal velocities, but the fraction of these falls as $\exp(-\beta^2 c'^2)$, and the overall validity of the distribution may be assessed on the assumption that the thermal speeds are of order $1/\beta$.

Additional terms in the Sonine polynomial expansions may be included to determine higher order approximations. Chapman and Cowling (1952) show that, in the second approximation, the coefficient of viscosity is increased by the factor $3(\eta-5)^2/\{2(\eta-1)(101\eta-113)\}$ and the coefficient of heat conduction by the factor $(\eta-5)^2/\{4(\eta-1)(11\eta-13)\}$. These are simple numerical constants that range from 0 in each case for Maxwell molecules to 0.0149 and 0.0227 for hard sphere molecules. However, additional terms are added in the second approximation to the distribution function and the numerical changes to this quantity are much greater. For example, consider a stationary gas with a temperature gradient in the y-direction. The first approximation is obtained by setting the final term in eqn (3.87) to zero. It may be shown that the second approximation is

$$f = f_0\Big(1 - \frac{4}{5}\frac{1}{45\eta^2-106\eta+77}\{-2(\eta-5)(\eta-1)\beta^4 c'^4 + (59\eta^2-190\eta+147)\beta^2 c'^2 - 10(13\eta^2-37\eta+28)\}\frac{K\beta^2 v'}{nk}\frac{\partial(\ln T)}{\partial y}\Big), \tag{3.88}$$

where K now represents the second approximation including the above-mentioned numerical factor.

The viscous stress tensor and heat flux vector in a monatomic gas become, in the subscript notation,

$$\tau_{ij} = \mu\Big(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\Big) - \frac{2}{3}\delta_{ij}\mu\frac{\partial u_k}{\partial x_k}, \tag{3.89}$$

and

$$q_i = -K\frac{\partial T}{\partial x_i}, \tag{3.90}$$

respectively. The substitution of these relationships into the conservation equations makes them a determinate set and leads to the monatomic gas form of the *Navier–Stokes equations* of continuum gas dynamics.

Because the derivation of the above expressions for the transport properties has been based on only the first term in the Chapman–Enskog expansion of eqn (3.51), there are limits on the validity of the Navier–Stokes equations, as discussed in §1.2. The evaluation of the next term in this expansion (which should not be confused with the Sonine expansion) leads to a set of very complicated higher-order continuum equations, called the *Burnett equations*. There is now growing evidence (Fiscko and Chapman, 1989) that the use of the Burnett equations extends the validity of the continuum model to flows that are more rarefied than those for which the Navier–Stokes equations are valid. Moreover, it has been shown (Pham-Van-Diep et al. 1991) that, even when the local Knudsen number is such that the Navier–Stokes equations would be expected to lead to an 'exact' solution, the Burnett equations can provide a more accurate description of a gas flow.

The diffusion effects in a binary gas mixture are best explained through the diffusion equation. The diffusion velocity for a single component of the mixture was defined by eqn (1.45), and the *relative diffusion velocity* may be written (Chapman and Cowling 1970),

$$\boldsymbol{C}_1-\boldsymbol{C}_2=\frac{-n^2}{n_1n_2}D_{12}\Big(\nabla(n_1/n)+\frac{n_1n_2(m_2-m_1)}{n\rho}\nabla(\ln p) - \frac{\rho_1\rho_2}{p\rho}(\boldsymbol{F}_1-\boldsymbol{F}_2)+k_T\nabla\ln T\Big). \tag{3.91}$$

There are four terms within the large parentheses on the right-hand side of this equation. The first involves the *concentration gradient* in the gas and the diffusion acts to reduce this gradient. The other three terms are due to effects that act to set up concentration gradients. The second term is for *pressure diffusion* which causes heavy and light gases to move towards regions of high and low pressure, respectively. The third, or *forced diffusion*, term is present only when there is a force $\boldsymbol{F}$ per unit mass that acts differentially on the molecules of the two species. The final term is due to *thermal diffusion* and introduces the thermal diffusion ratio k_T. Thermal diffusion is a subtle effect that depends on the differences in molecular size, and was discovered only in response to its prediction by the Chapman–Enskog theory. It causes the larger and (generally) heavier molecules to migrate towards the cooler regions of a flow. Pressure diffusion and thermal diffusion act in opposite directions in an expansion. However, they act in the same direction in the stagnation region of the hypersonic flow over a cold blunt body and the separation can be significant in the flow of a partly dissociated gas which necessarily involves large mass ratios. Forced diffusion can generally be neglected in neutral gases.

The Navier–Stokes equations are generally extended to include the concentration gradient effects that reduce concentration gradients when these are present as a result of boundary conditions or chemical reactions. However, the Navier–Stokes formulation almost always neglects the pressure and thermal diffusion effects that act to introduce concentration gradients. These are generated even when the boundaries involve a uniform composition and there are no chemical reactions. These separation effects can be significant in, for example, hypersonic gas flows and the omission of these terms can lead to significant errors.

As noted earlier, the correction to the first approximation to the diffusion coefficient due to the inclusion of extra terms in the Sonine polynomial expansion is much larger than the corresponding corrections to the viscosity and thermal conduction coefficients. As discussed by Chapman and Cowling (1970), the correction can be as high as 13%. Moreover, the corrections are not just numerical factors, but introduce the species number ratio or concentration as an additional parameter on which the diffusion coefficients depend. This latter point introduces additional difficulties to the continuum formulation through the Navier–Stokes equations.

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